Ziphozihle Maop

ch4

45

98%

TOOLS, EQUIPMENT AND MATERIALS

SUMMATIVE ASSESMENT

EASTC TECHNOCENTRIC VARSITY

Module #	671101000-KM-02
NQF Level	level 4
Notional hours	80
Credit(s)	Cr 8
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

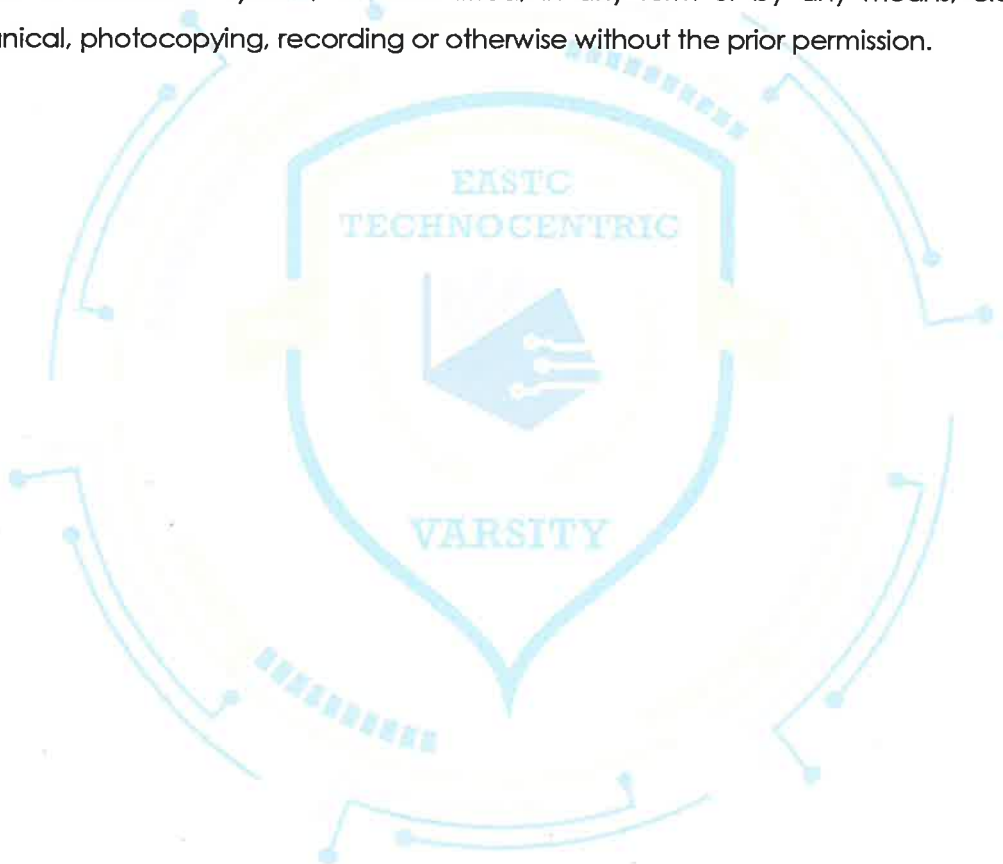
Details	Please Complete this Section
Name	Ziphozihle Maopi
Contact Address	
Telephone (H)	N/A
Telephone (W)	N/A

Z

Facsimile	N.A
Cellular	084 045 6528
E-mail	mappizizipho@gmail.com

Copyright

All rights reserved. The copyright of this document, its previous editions and any annexures thereto, is protected and expressly reserved. No part of this document may be reproduced, stored in a retrievable system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise without the prior permission.



Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

Z

LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	

Z

Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1st Reassessment	Assessors summary report / instruments	Learner file Date:	
2nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: Ziphozihle Maopi	Manager:
Trainer:	Mentor / Coach:
Assessor: Balenagh Jabadi	Moderator:

Z

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Zipnozinle Maqpi

Learner's signature: _____ Date: _____

Assessors Name: Balenah Tyabadi

Assessor's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
   - If the laid down assessment procedures were not followed during assessments
   - If not all evidence available was taken into account during the assessment
   - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
   - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
   - Not all the range items were available for assessment

2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.

Z

3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal "form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.

4. Should the internal moderator re-affirm the assessor' decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

EASTC TECHNOCENTRIC VARSITY

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Z

Name of employee assessed: _____

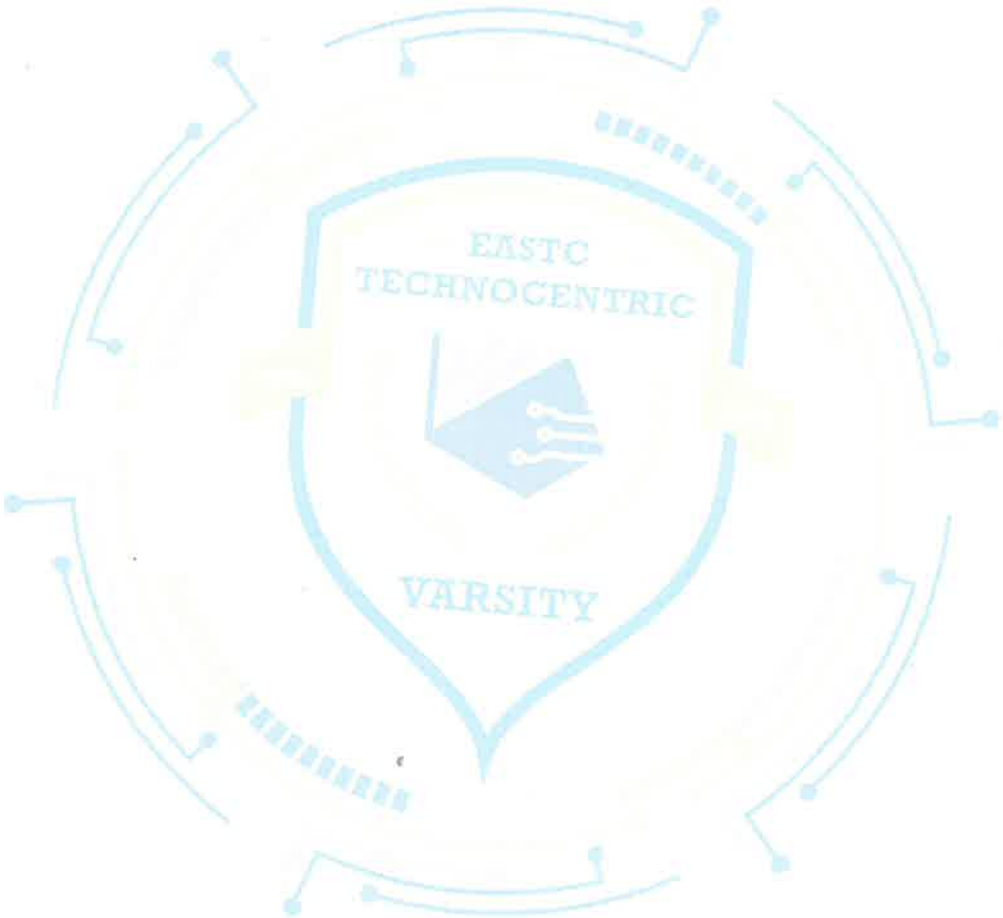
Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Z


EASTC
TECHNOCENTRIC
VARSITY

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

EASTC TECHNOCENTRIC VARSITY

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate **and put them at ease**	✓
2	**Explain the purpose of the meeting** (why you are there and how long the meeting will take)	✓
3	**Explain the** ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	**Explain** ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	**Explain the assessment process?** ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	**Give Learner copies of the following documentation and explain each document** ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	**Discuss the assessment plan** (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	**Tell the candidate his/her rights and responsibilities, the assessment procedures and policies** ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	**Ensure the assessment environment is appropriate** or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Z

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Ziphozihle Ngopi	Signature
Assessors Name: Balcronah Yqbadi	Signature:

EASTC TECHNOCENTRIC VARSITY

INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

EASTC TECHNOCENTRIC VARSITY

LEARNING UNIT 1:

KM-02-KT01: Hand tools and power tools (50%)

Learning Outcome 1:	KT0101 Types, uses and care of hand tools
Learning Outcome 2:	KT0102 Types, uses and care of portable power tools
Learning Outcome 3:	KT0103 Types, uses and care of fixed power tools

Task	Questions Description	Marks
1	Identify hand tools and describe their uses	10

Pliers: There are 2 types of pliers insulated and regular pliers. Electricians utilise pliers for a variety of tasks including gripping wires securely, cutting wires

Wire Strippers: Wire strippers are essential tools for any electrical, designed to remove insulation from wires without damaging the copper conductor inside.

Cable Cutters: Are used for cutting thick electrical cables cables and armored wires. Many cable cutters also have insulated handles for safety and ratcheting mechanisms that make cutting easier on your hands.

Utility knife: A sharp utility knife is used for cutting insulation, trimming wires, stripping cables, and also used for handling small tasks on-site.

Screwdriver: This tool is used to turn screws into wood or metal. It is a very handy tool ~~to use~~ and easy to use

Z

Crimping Tool: A crimping tool is a ~~hand~~ used for connecting wires together safely

Nut driver: Nut drivers help tighten nuts and bolts quickly and safely

Allen Wrench: This small tool is used to tighten bolts with a hexagonal shape

~~Hammer: A hammer is used to secure~~
Hammer: Standard claw hammers are used for mounting electrical boxes, securing fasteners, and general construction tasks

EMT Bender: Is used to bending conduit to fit installation requirements

13

LEARNING UNIT 2:

KM-02-KT02: Measuring and testing instruments (50%)

Learning Outcome 1:	KT0201 Portable electrical measuring instruments
Learning Outcome 2:	KT0202 Types of electrical measuring and testing instruments
Learning Outcome 3:	KT0203 Safe use of measuring and testing instruments
Learning Outcome 4:	KT0204 Construction and operating principles of measuring and testing instruments
Learning Outcome 5:	KT0205 Methods of connecting measuring and testing instruments in circuits
Learning Outcome 6:	KT0206 Applications and methods of using electrical measuring and testing instruments
Learning Outcome 7:	KT0207 Care and maintenance of single and three phase measuring and testing instruments
Learning Outcome 8:	KT0208 Instrument transformers
Learning Outcome 9:	KT0209 Types and functions of panel mounted electrical measuring and testing instruments
Learning Outcome 10:	KT0210 Safe use of panel mounted measuring and test instruments
Learning Outcome 11:	KT0211 Protection of measuring instruments when connected in a circuit.

Task	Questions Description	Marks
2	a) Identify the basic components of various measuring and test instruments and describe the basic principle of operation. b) Describe the applications and methods of using electrical measuring and testing instruments in direct and alternating current circuits.	10

Z

2a.)

1. Multimeter: A multimeter measures voltage, current, resistance by converting inputs into digital readings. It works on the principle of Ohm's law and uses internal circuits to convert analog signals to digital values.

2. Clamp Meter: Measures current via magnetic field detection around a conductor. The Hall effect sensor detects the magnetic field and converts it to a voltage proportional to the current.

3. Insulation Tester: Tests insulation resistance using high voltage, measures leakage current. The higher the resistance, the better the insulation.

4.) Oscilloscope: Visualises voltage waveforms over time for signal analysis. It shows the signal's amplitude, frequency and other characteristics.

5.) Earth loop tester: measures loop impedance to check earth fault protection efficacy. It simulates a fault and measures the loop resistance.

b.)

1) Multimeter

DC: Measures voltage, current, resistance, and continuity. It is used in series or parallel, ensuring correct range and polarity.

AC: Measures voltage, current with a clamp and frequency. Ensures AC setting and safety precautions.

Task	Questions Description	Marks
3	Describe safety and functionality checks to be performed on measuring and testing instruments before use	5

1.) Visual Inspection: Check for any physical damage ~~fere~~ like cracks, dents or corrosion) on the instrument and its accessories. Inspect cables and connection for wear, fraying or broken insulation.

2) Calibration Status: Verify if the instrument is properly calibrated and has a valid calibration certificate. Ensure that the calibration date falls within the acceptable range specified by the manufacturer.

3.) Power Supply: Ensure that the power supply (batteries, AC adapter) is functioning properly within the required specifications for the instrument. Check for proper grounding or isolation, especially for electrical measuring devices.

4.) Safety Features: Confirm that safety features like fuses, circuit breakes are intact and functional. Ensure that protective covers and guards are in place; especially for high-voltage instruments

5.) Environmental Conditions: Assess the work area for compliance with the instrument's enviromental requirements. Ensure that the are is free from pontential hazards or obtacles. This will ensure your measuring instruments are both safe and functional, ultimately aiding in the ~~reliabity~~ realiability of results obtained.

5

Z

Task	Questions Description	Marks
4	a) Describe correct methods of handling and storing measuring and testing instruments b) Describe, with the aid of drawings, the connection of measuring instruments to single phase and three phase circuits	10

4a)

1) Proper Handling Techniques: Training and education are crucial for all personnel. Avoid physical shock by handling instruments gently and using protective gear. Keep instruments away from extreme temperatures, humidity, dust and vibrations.

2) Storage Conditions: store instruments in a climate-controlled environment to minimize exposure to humidity, dust and temperature fluctuations. Use dry cabinets for sensitive instruments and regularly clean storage areas

3) Battery management: Replace batteries as needed to ensure reliable operation. Remove batteries if the instrument will not be used for an extended period to prevent corrosion

4) Cleaning: When cleaning electrical measuring and testing instruments use a soft, dry cloth to clean exterior surfaces, avoid solvents

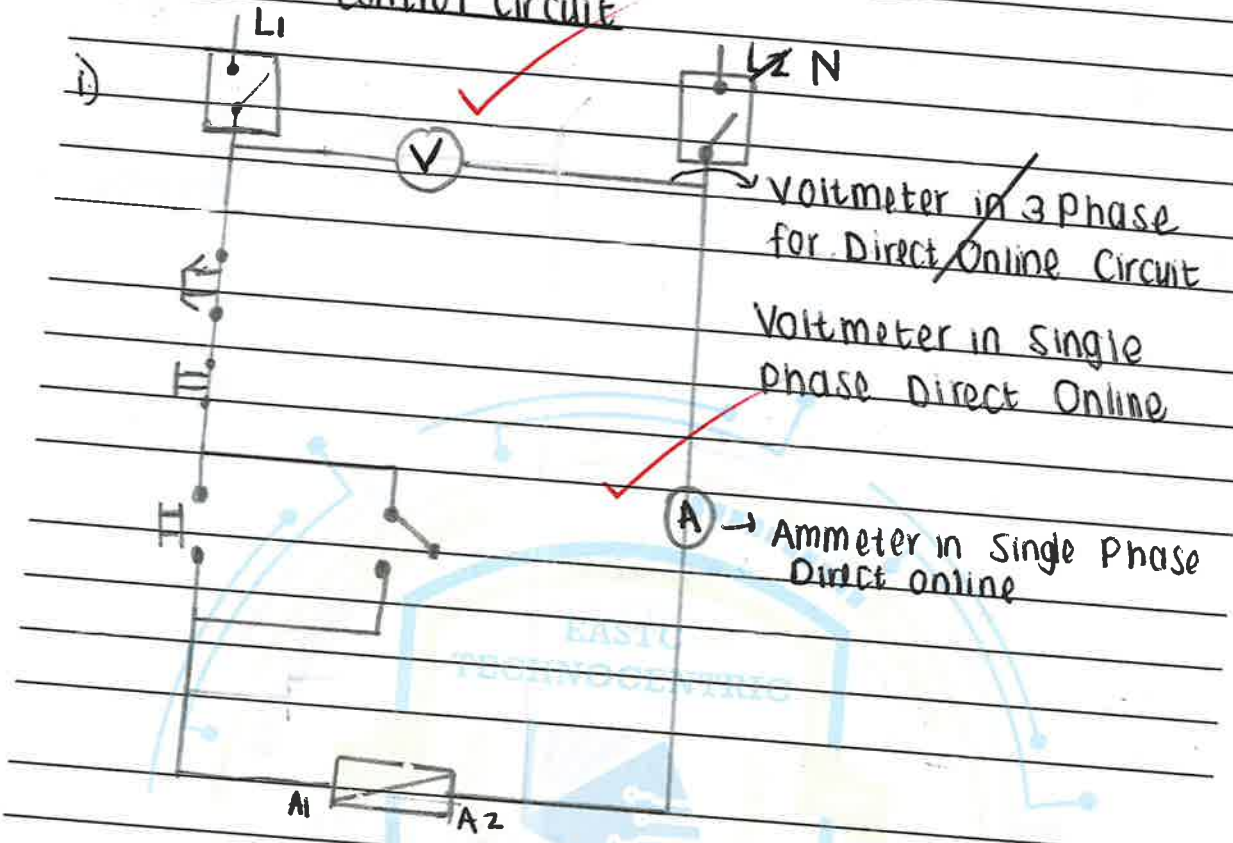
5) Inspection: Check leads and probes of electrical measuring and testing instruments for wear or damage before use.

9

Z

b) Control Circuit

i)



Voltmeter in 3 Phase for Direct Online Circuit

Voltmeter in Single Phase Direct Online

A → Ammeter in Single Phase Direct online

2.) Main Circuit Single Phase

L N

2) Main Circuit 3 Phase Foward / Reverse

The ammeter in the main circuit of 3 Phase Foward/Reverse measures the flow of electric charge in a circuit showing how much current passe through a conductor per second

The voltmeter in 3Phase Foward/Reverse measures the potential difference between two points in the ~~dia~~ circuit digram.

Z

and Radio-frequency interference

6) Grounding: Proper grounding techniques also called Earthing provide a path for fault current and protects against electric shock. Grounding prevents ground loops and ensure safe operation

7.) Surge arrestor: Also protects against voltage spikes preventing instruments or appliances from being highly damaged.

8.) Single and Double Diode Protection: Using diodes can help divert excess current away from meter, protecting it from damage due to overcurrent conditions

9.) Protective Relays: These devices detect abnormal conditions in the circuit and trigger disconnection. Abnormal conditions include overcurrent, overvoltage, under-frequency or short circuit. This device trips the circuit breaker to prevent damage

10.) Proper Wiring: Using appropriate wire sizes and ensuring proper connections are crucial for safety. Oversized wires can handle higher current, reducing the risk of overheating and fire hazards

10

Z

Task	Questions Description	Marks
5	Describe the various methods of protection when connecting measuring instruments in circuits	10

~~1) Fuses~~

1) Fuses and Circuit Breakers: Fuses contain a thin wire that melts when current exceeds a safe limit, interrupting the circuit and protecting the instrument from damage. Circuit Breakers perform a similar function but can be reset after tripping.

2) Current-limiting Resistors: These resistors are connected in series with the instrument to restrict the current flow, ensuring that even under fault conditions, the current remains within safe operating limits

~~3) Diode Protection: Sing~~

3) Isolation: Optical isolation transformers, or galvanic isolation techniques electrically separate the instrument from the circuit, protecting against high voltages or ground loops.

4) Input protection: Many instruments have built-in input protection such as clamping diodes or resistor networks, to protect against ~~overload~~ overvoltage or overcurrent

5) Shielding: Shielded cables or enclosures protect the instrument from electromagnetic interference

Assessor's Report

Candidate's Name	Ziphozihle Maqopi	ID No.	0802030118086
Assessor's Name	Baiennah Jabadi	Reg. No.	Namb-AI 2025/5307
Title			

ASSESSMENT DECISION

Outcome	C	NYC	Comments

OVERALL COMPETENCE JUDGEMENT:

Comments ~~Excellent~~

Date	
~~7-8~~	
Signature of Assessor	Signature of Candidate

Z

Candidate Feedback Report

Candidate's Name	Ziphozihle Maqopi	ID No.	0802030118086
Assessor's Name	Baienah Jabadi	Reg. No.	NAMB-A1 2025/5307
Title	Tools, Equipment and Materials		

ASSESSMENT DECISION

Source of Evidence	C	NYC	Comment
Knowledge questions	✓		Good
Product /Work sample	✓		Good
Project	✓		Good
Indirect Evidence	✓		Good
Overall Assessment Decision			
Additional Notes			

Date	
Signature of Assessor	Signature of Candidate

Candidate Appeal Form

Candidate's Name:		ID No.	
Assessor's Name:		Reg. No.	
Title:			
Date:			
SECTION 1			
Candidate's reason for disagreeing with the assessment decision			
Assessor's rationale for the assessment decision			
Candidate's signature			
Assessor's signature			
SECTION 2			
Moderator's decision			
Moderator's signature			

EASTC TECHNOCENTRIC VARSITY

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name	Balenmah Tjabadi	Reg. No.	Namb-A/2025/5307
Candidate's Name	Ziphozihle Mqopi	ID No.	0802030118086
Title	Tools, Equipment and Materials		

MODERATION DECISION

Outcome	C	NYC	Comments
Overall Moderation Decision			

Feedback to Assessor	
Action Required	
Date of Moderation	
Signature of Moderator	
Signature of Assessor	
Signature of Candidate	

ASSESSMENT REVIEW

NAME of LEARNER Zipnozinhle Mapi	**NAME of ASSESSOR** Baiennah Tjabadi
VENUE EASTC TECHNOCENTRIC VARSITY	**DATE of REVIEW**

Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment was practical?	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The venue and equipment was functional?	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
An opportunity to appeal was given	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The evidence was recorded	☒ Agree ☐ Disagree	☑ Agree ☐ Disagree	

LEARNER'S DECLARATION OF UNDERSTANDING

I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid

Learner Date	Assessor Date	Moderator Date